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BluetoothDetective Proposal

System Description

BluetoothDetective is a mobile attachment device which is used to locate and display information about Bluetooth signal emitting mobile devices in the proximity of the user. The BluetoothDetective is to be connected to a cell phone through a USB. The structure of the device is an Aduino board connected to 4 Bluetooth modules neatly encompassed in a plastic box. The box would be about the same shape and size as a standard SSD, and on the outside it will have a cell phone holder so the device can be physically attached to the phone and easily held alongside it as a single unit. The cell phone holder will be adjustable to accommodate various cell phone sizes. The device detects signal strength of different Bluetooth sources. This information is transferred to the cell phone. The BluetoothDetective app (installed by the user on their phone) determines where the source is located in the space. The app provides the camera view to the user. Through the camera view, the app draws a box around Bluetooth devices detected by the BluetoothDetective. The app displays the following information: distance of the detected device, Bluetooth address and other information that can be detected. The app would allow options of tracking devices based on their distance or Bluetooth address. Additionally the user could choose which information will be displayed regarding the detected source.

Target Users

The use cases in which the device is mainly envisioned only include a primary user (for example, any person using the device to find their lost Bluetooth earphones). Cases where secondary users are involved are more uncommon.

An example would be a mom who is familiar with using Bluetooth devices. She can use BluetoothDetective to look for any bluetooth devices her family misplaces. A secondary user could be her 7 year old son who doesn't have his own phone, but is prone to losing Bluetooth devices. Her son likes playing games on the family’s ps4, but sometimes he loses the Dualshock controller for it, and needs his mom’s help to find it. He cares that the device is easy to use and reliable, so he can continue playing his video games.

Another example could be a government agent using BluetoothDetective to ensure that the room he is about to have a meeting in is not secretly wired or monitored by Bluetooth microphones or cameras. The secondary user is the government agency for which the agent works. It cares that the device is easy to use and reliable because that ensures confidentiality of the meeting about to take place.

Similar Products

There are several types of products similar to BluetoothDetective. One of such products is an EMF meter, which is a device used to detect radiation waves (including Bluetooth signals) coming from objects at a certain distance. The objects it may detect are power lines, cell phones, smart meters, microwaves, cell towers, static, among others ("EMF Meter,Advanced GQ EMF-390 Multi-Field Electromagnetic ...", n.d.). BluetoothDetective differs from the EMF meter in its purpose, which is to detect devices emitting Bluetooth signals. The BluetoothDetective is attached to a cell phone, and allows the user to utilize the app with a camera view, which provides information about the detective device. The EMF meter does not provide the camera view to the user ("EMF Meter,Advanced GQ EMF-390 Multi-Field Electromagnetic ...", n.d.).

BLE Finder is a mobile app which is used to help users find lost Bluetooth devices. The app works by detecting signal strength and displaying it to the user. The user is supposed to use signal strength as feedback on whether they are moving closer or further away from the device. ("Find My Lost Bluetooth Device", 2020) The difference between LightBlue and BluetoothDetective (apart from the fact that BluetoothDetective is an actual device + app system) is that BluetoothDetective determines where in space relative to the user the device is located, and on top of that it gives the user a neat visualization in the form of a box drawn around the source of the signal in the camera view.

Another product, the Tile app, is a similar product to BluetoothDetective in that it also uses bluetooth to find missing belongings. However, the difference is that the Tile app also requires an accompanying bluetooth device (called a "Tile") to be attached to a belonging, and it is used to find any kind of belonging - not just bluetooth devices - as long as a "Tile" is attached to it. BluetoothDetective does not require any objects it can detect to have attachments.

Visualization

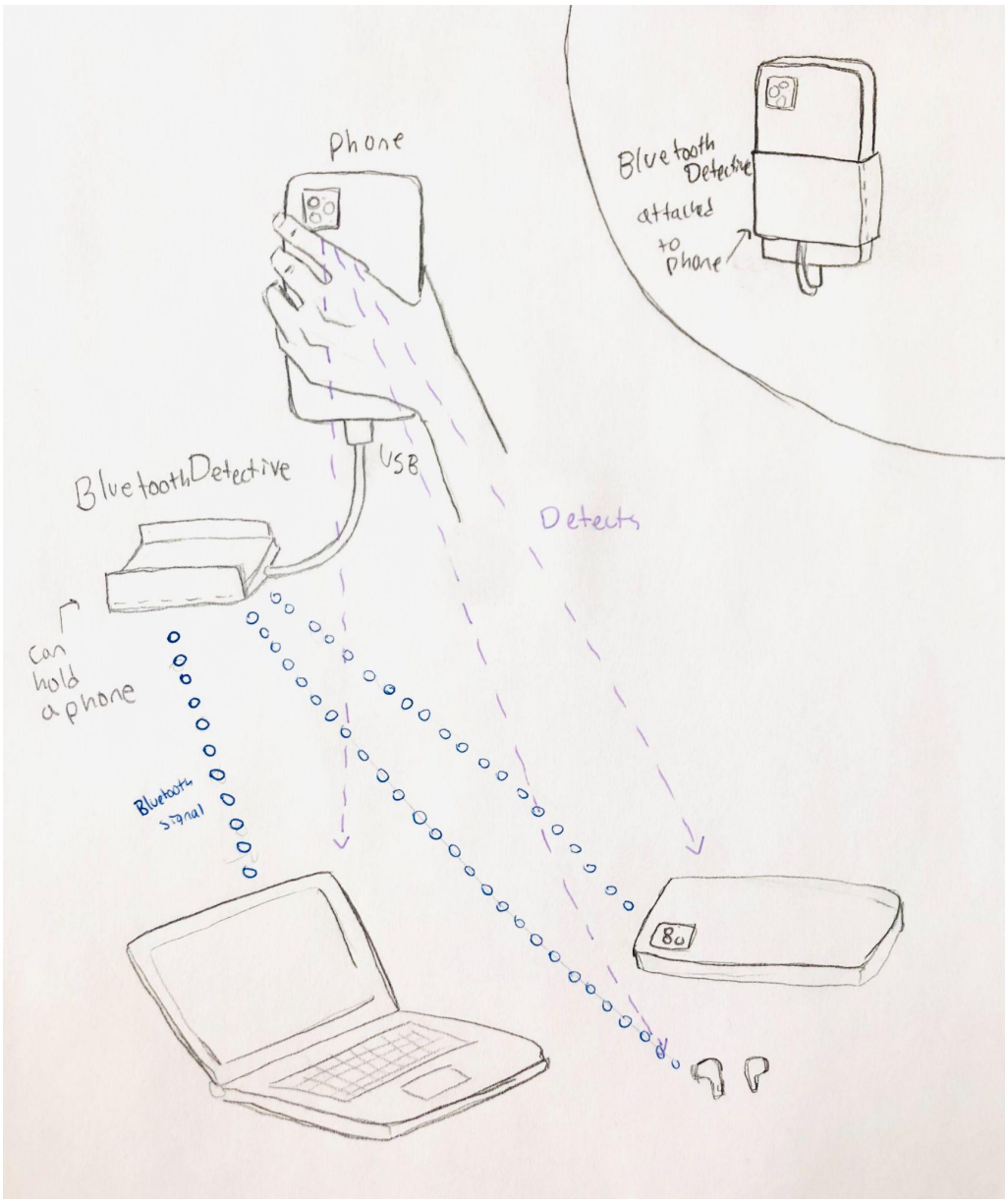


Figure 1: A sketch of the BluetoothDetective general form and functionality. The device is connected to a phone through a USB, and allows the phone camera to display the Bluetooth connected devices. BluetoothDetective may also be attached to the phone as displayed in the top right-hand corner.

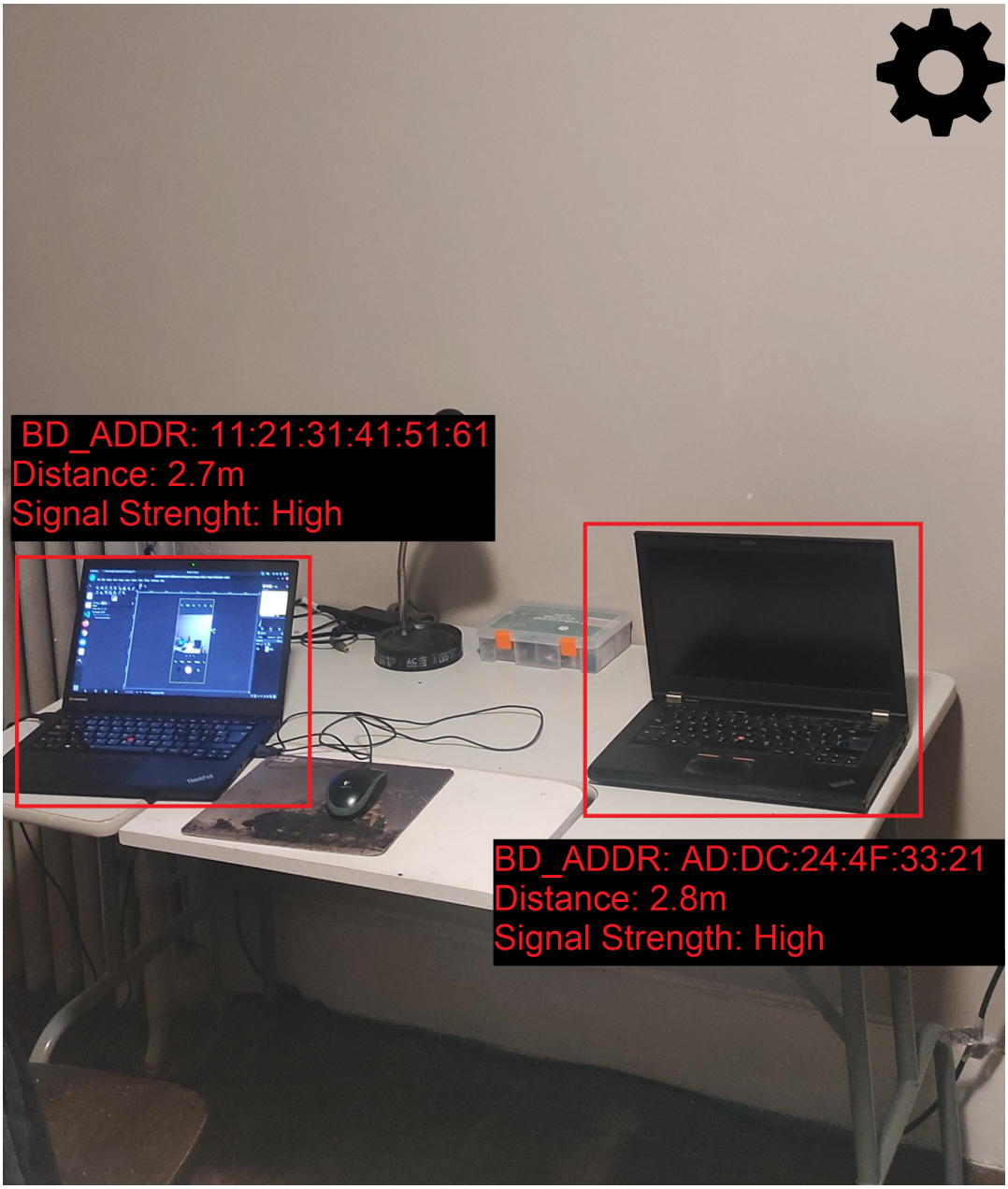


Figure 2: A rendition of how the camera view would look like. Red boxes would be drawn around devices which are detected, with some information displayed next to the devices such as the device's MAC address and distance from the user. There is a settings button in the top right corner for adjusting which information is shown and which devices would be tracked.

Prototypes

Physically the device will consist of an Arduino Uno board connected to four Bluetooth Modules (HiLetgo HC-05). The Arduino/Bluetooth system will be encompassed by a 3D printed case. Outside of the case will have a phone holder. The smartphone will be connected to the Arduino via cable. For prototyping we will use an Android Smartphone. The app on the phone will be developed via Android Studio.

Sources

"EMF Meter,Advanced GQ EMF-390 Multi-Field Electromagnetic ..." Amazon. Amazon. Accessed February 16, 2022. https://www.amazon.com/Advanced-GQ-Multi-Field-Electromagnetic-Radiation/dp/B07JGJ897T.

"Find My Lost Bluetooth Device." App Store., February 1, 2020. https://apps.apple.com/us/app/find-my-lost-bluetooth-device/id1488361557.

"Tile's Bluetooth Tracker Devices Can Find Just about Anything You're Tracking: Tile." Find Your Keys, Wallet & Phone with Tile's App and Bluetooth Tracker Device. Accessed February 16, 2022. https://www.thetileapp.com/en-us/products?msclkid=9716f7eb3e9017b6264095e6d09992aa&utm_source=bing&utm_medium=cpc&utm_campaign=%28P%3AG%29+%28CT%3APROS%29+%28GEO%3AUS%29+%28CA%3ANB-precise%29+%28N%3AGSN%29+%28AT%3AKWs%29+%28AU%3AKWs-exact%29&utm_term=bluetooth+locator&utm_content=bluetooth+locator.